Day 3 — Naming the 15 Tithis

Module: Moon Phases & Tithi · **Band:** Middle · **Time:** 45 min

Learning objective

Students will define *tithi* in terms of the Moon's elongation from the Sun (12°), recite the 15 tithi names of a pakṣa in order, and explain why a tithi need not equal a 24-hour calendar day.

Materials

- Board with three columns ready: Number | IAST name | English (ordinal)
- A printed handout of the 15 tithi names (one per student)
- The white ball and torch from Day 1 (one quick re-use)
- Notebooks

Vocabulary introduced today

- *tithi* (IAST: tithi; lit. "lunar day") — 1/30 of a synodic month; precisely, the time the Moon takes to gain 12° of angular distance ("elongation") from the Sun.
- *elongation* — the angle between the Sun and Moon as seen from Earth (0° at *amāvasyā*; 180° at *pūrṇimā*).

Lesson flow

Warm-up (5 min)

- Daily prompt: *"Did you see the Moon last night? What pakṣa is it?"* — 2 students share.
- Recall: write on the board — "29.5 days, 2 pakṣas, 2 anchor days." Ask the class to fill in: how many days in a pakṣa? (~14.75.) How many parts will we divide that into? (15.)

Core (20 min)

1. **The tithi as a unit.**

   "A tithi is one-thirtieth of a full lunar cycle. So a synodic month has 30 tithis. Each pakṣa has 15."

   On the board:

   ```
   Synodic month (29.53 days)
        ├── Śukla pakṣa = 15 tithis (1 → 15)
        └── Kṛṣṇa pakṣa = 15 tithis (1 → 15)
   ```

2. **The precise definition (this is important).**

   "A tithi is NOT one-thirtieth of a day. It is one-thirtieth of the cycle."

   The full cycle covers 360° of elongation (the Moon goes all the way around). Divide 360° by 30 — each tithi is 12° of elongation.

One tithi = the time during which the Moon gains 12° of elongation on the Sun.

On average, this takes about 23 hours and 37 minutes. But the Moon doesn't move uniformly — sometimes faster (perigee, closer to Earth) and sometimes slower (apogee, farther). So a single tithi can last anywhere from ~21.5 hours to ~26 hours.

3. **Why a tithi is not a day — three consequences.**
   - Some tithis "skip" a sunrise. If a tithi starts at 7 a.m. and ends at 5 a.m. the next morning, it never overlaps with a sunrise. This is called *kṣaya* tithi ("decayed" — it never had a sunrise of its own).
   - Some tithis "span" two sunrises. If a tithi starts at 5 a.m. on one day and ends at 7 a.m. two days later, it covers two sunrises. This is *vṛddhi* ("growth").
   - Most tithis are tagged to one calendar date — the date during which the tithi is current at sunrise.
4. **The 15 names — they are Sanskrit ordinals.**

   On the board, build the list. Students copy. (Show *śukla* column; same names apply to *kṛṣṇa* with the prefix.)

#	IAST	English (ordinal)
1	*pratipadā*	"first"
2	*dvitīyā*	"second"
3	*tṛtīyā*	"third"
4	*caturthī*	"fourth"
5	*pañcamī*	"fifth"
6	*ṣaṣṭhī*	"sixth"
7	*saptamī*	"seventh"
8	*aṣṭamī*	"eighth"
9	*navamī*	"ninth"
10	*daśamī*	"tenth"
11	*ekādaśī*	"eleventh"
12	*dvādaśī*	"twelfth"
13	*trayodaśī*	"thirteenth"
14	*caturdaśī*	"fourteenth"
15	*pūrṇimā* (or *amāvasyā*)	full / new moon

Note the pattern: 1–10 are simple ordinals; 11–14 are compound (*eka+daśī*, *dvā+daśī*, *trayaḥ+daśī*, *catur+daśī*); the 15th is always the anchor day — *pūrṇimā* at the end of *śukla*, *amāvasyā* at the end of *kṛṣṇa*.

5. **Sanskrit number recall.** *Eka, dvi, tri, catur, pañca, ṣaṣ, sapta, aṣṭa, nava, daśa* — count it together once. The tithi names will stick faster if students see them as ordinals first, Sanskrit terms second.

Activity (15 min) — Tithi drill

Two parts:

Part A (8 min) — paired recall. In pairs, partners alternate: one says a tithi number (1–15), the other names the tithi in IAST. Switch roles every minute. Goal: name all 15 in under 60 seconds by end of activity.

Part B (7 min) — date-to-tithi. Distribute a printed list of 5 dates and the tithi at sunrise on each (lookup done by the teacher from any standard *pañcāṅga* — e.g. for a sample week). Students answer: - *Is this date in śukla or kṛṣṇa pakṣa? - What's the Sanskrit name of the tithi? - What shape will the Moon be that night?*

Walk around. Check that students are reading the table, not guessing.

Wrap-up (5 min)

- **Formative quiz on a slip of paper (5 questions, 3 min)** — see `quizzes/formative.md` Part A.
- Collect for grading. Next day's class includes the result.
- **Preview Day 4:** *"Tomorrow we build a tithi calendar for THIS calendar month."*

Student-facing content

> **A tithi is the time the Moon takes to gain 12° on the Sun.**
>
> - 360° / 30 = 12° per tithi.
> - Average tithi length: ~23 h 37 min — close to a day but not exactly a day.
> - Some tithis skip a sunrise (*kṣaya*); some span two (*vṛddhi*).
>
> Each pakṣa has 15 tithis named as Sanskrit ordinals:
>
> *pratipadā, dvitīyā, tṛtīyā, caturthī, pañcamī, ṣaṣṭhī, saptamī, aṣṭamī, navamī, daśamī, ekādaśī, dvādaśī, trayodaśī, caturdaśī, pūrṇimā* (or *amāvasyā*).

Homework

- Memorise the 15 tithi names. Quiz yourself or be quizzed at home.
- Continue the Moon diary — now naming the tithi as well as the pakṣa.

Differentiation

- **One tier up:** Look up the formula for tithi computation: `tithi number = floor( ((Moon_ecliptic_longitude − Sun_ecliptic_longitude) mod 360) / 12 ) + 1`. Use an astronomy app (Stellarium is free) to get the longitudes at any moment and compute the tithi yourself. Compare with a published *pañcāṅga*.
- **One tier down:** Memorise only the 6 anchor tithis — *pratipadā, pañcamī, aṣṭamī, ekādaśī, caturdaśī, pūrṇimā/amāvasyā*. Fill the rest in next week.

Teacher notes

- The 12°-per-tithi rule is the single most important computational fact in this module. Drill it.
- Watch for "tithi = day" confusion — every quiz will catch this. The simplest fix: ask "if a tithi started at 7 a.m. and the next started at 5 a.m. the next morning, how many hours was that tithi?" (22.) Repeat with different times until students see that tithi length varies.
- Pronunciation: *ṣaṣṭhī* is hard — "shash-tee." *Ekādaśī* — "ay-KAH-da-shee." Don't rush.

Citation(s) used in this lesson

- Paraphrase from *Sūrya-siddhānta*, chapter 2: the tithi is defined as the period during which the Moon and Sun separate by 12° of arc. Cited by text name only; we do not quote a specific verse.